Business Intelligence Internship Assignment
Data Simulation & Analytics

Anuj Sharma

1 Introduction

This assignment simulates the car insurance policy portfolio of an insurance company and analyzes claim patterns and profitability. A synthetic dataset was generated to represent policy sales and claims activity. The analysis focuses on premium collection, claim trends, and loss ratios to evaluate the overall performance of the insurance portfolio.

2 Dataset Overview

The simulated datasets contain the following structure:

- Policy Sales Dataset: **1,000,000 records with 8 attributes**
- Claims Dataset: **49,274 records with 6 attributes**

These datasets represent simulated insurance policy purchases and claim events.

3 Approach

Policy purchase dates were evenly distributed between January 1, 2024 and December 31, 2024. The policy start date was calculated as 365 days after the purchase date, and the policy end date was determined based on the policy tenure.

In 2025, claims were generated only for vehicles purchased on the 7th, 14th, 21st, and 28th of each month. 30% of these vehicles filed a claim on the first day of the policy start date.

In the period January–February 2026, 10% of vehicles with a 4-year policy tenure filed a claim, distributed evenly across the 59-day period.

Python was used to generate datasets and perform analytical queries.

4 Assumptions

Assumption	Description
Vehicle Value	100,000 assumed for every vehicle
Premium	100 per year of policy tenure
Claim Amount	10% of vehicle value (10,000)
Policy Distribution	Policies evenly distributed across the year 2024
Claim Rule	Each vehicle can file only one claim per year

5 Total Premium Collected

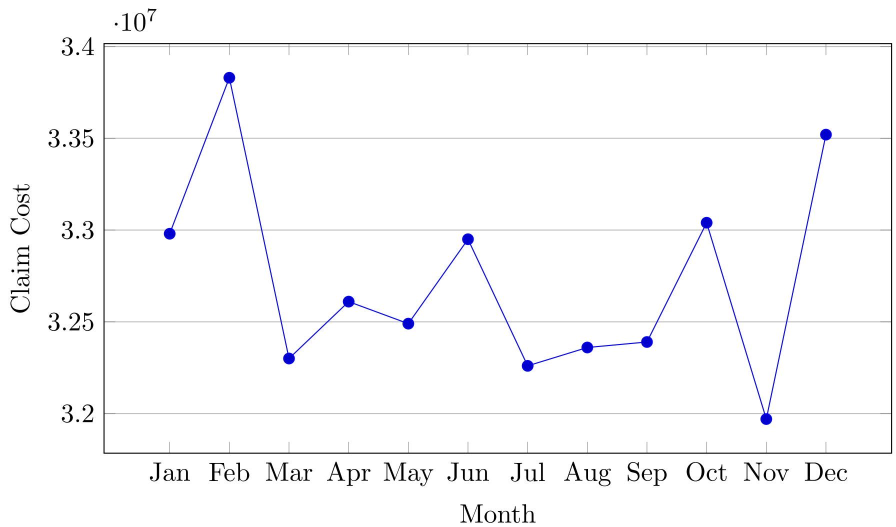
The total premium collected from all simulated policies sold during the year 2024 is:

240,065,600

6 Claim Cost by Year and Month

Year	Month	Claim Cost
2025	January	32,980,000
2025	February	33,830,000
2025	March	32,300,000
2025	April	32,610,000
2025	May	32,490,000
2025	June	32,950,000
2025	July	32,260,000
2025	August	32,360,000
2025	September	32,390,000
2025	October	33,040,000
2025	November	31,970,000
2025	December	33,520,000

7 Visualization: Monthly Claim Cost Trend

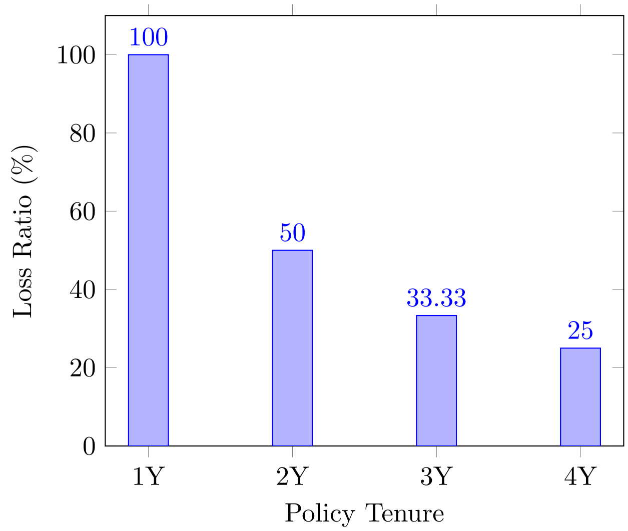


This chart shows the monthly claim cost trend for the year 2025.

8 Claim to Premium Ratio by Policy Tenure

Policy Tenure	Total Claim	Total Premium	Loss Ratio
1 Year	78,820,000	788,200	100%
2 Years	118,230,000	2,364,600	50%
3 Years	156,100,000	4,683,000	33.33%
4 Years	139,590,000	5,583,600	25%

9 Visualization: Loss Ratio by Policy Tenure



The visualization shows that loss ratio decreases as policy tenure increases, indicating higher profitability for longer-term policies.

10 Loss Ratio by Policy Purchase Month

Purchase Month	Claim Cost	Premium	Loss Ratio (%)
January	41,540,000	1,127,600	36.84
February	41,360,000	1,108,900	37.30
March	41,030,000	1,123,600	36.52
April	40,530,000	1,101,700	36.79
May	41,470,000	1,144,300	36.24
June	41,390,000	1,130,100	36.63
July	40,410,000	1,095,500	36.89
August	41,160,000	1,135,700	36.24
September	40,540,000	1,104,000	36.72
October	41,790,000	1,137,800	36.73
November	40,000,000	1,095,300	36.52
December	41,520,000	1,114,900	37.24

The loss ratio across purchase months remains relatively stable between 36% and 37%.

11 Portfolio Loss Ratio

The estimated portfolio loss ratio (Total Claims ÷ Total Premium) for the simulated dataset is approximately:

36%

12 Estimated Future Claim Liability

Assuming that all vehicles which have not yet filed a claim eventually file exactly one claim during the remaining policy tenure, the estimated future claim liability is:

9,511,330,000

13 Premium Earned till February 28, 2026

Based on the daily premium recognition method, the premium earned by the insurer up to February 28, 2026 is:

66,214,818.36

14 Key Insights

- Total premium collected from the simulated portfolio is approximately 240 million.
- Claim costs remain relatively stable across months, indicating consistent claim behavior.
- Loss ratio decreases significantly as policy tenure increases.
- The 4-year policy tenure is the most profitable segment with the lowest loss ratio.
- Estimated future claim liability is approximately 9.51 billion, highlighting the importance of maintaining sufficient reserves.